

Exercise 7: Financial Forecasting

Scenario:

You are developing a financial forecasting tool that predicts future values based on past data using System;

```
class Program
{
    static double FinancialForecasting(int currentValue, int year, double avgGrowthRate)
    {
        if (year == 0)
            return currentValue;

        return FinancialForecasting(currentValue, year - 1, avgGrowthRate) + avgGrowthRate;
    }

    static void Main()
    {
        int[] pastData = { 1000, 1200, 1400, 1600 };

        int totalGrowthRate = 0;

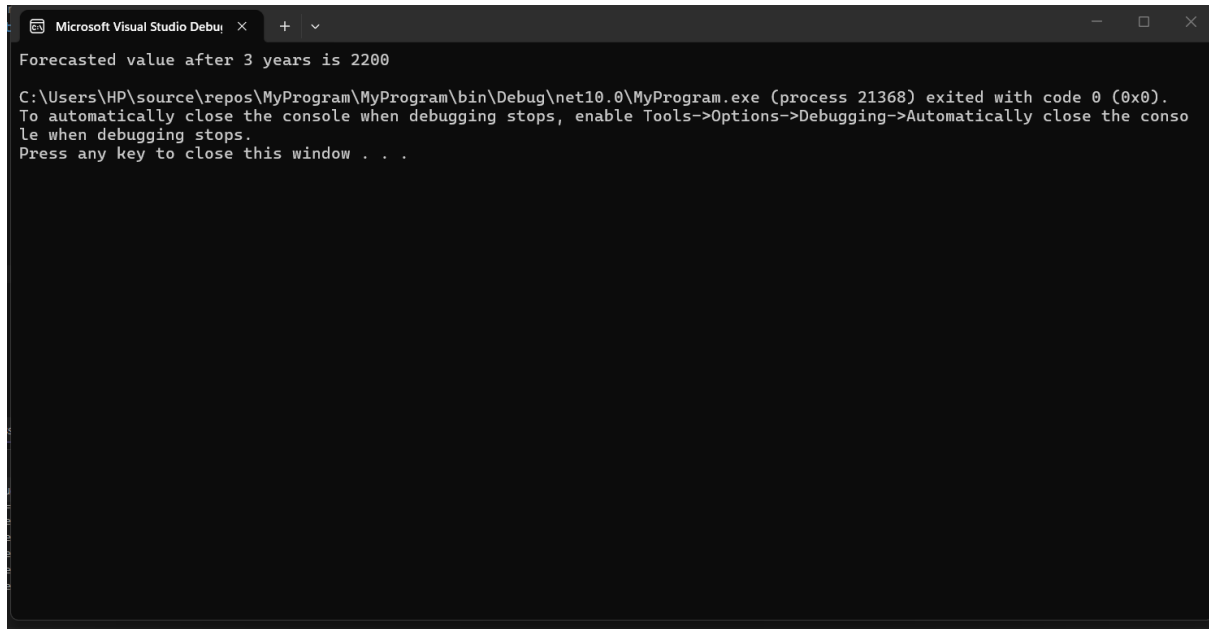
        for (int i = 0; i < 3; i++)
        {
            totalGrowthRate += pastData[i + 1] - pastData[i];
        }

        double avgGrowthRate = totalGrowthRate / 3.0;

        int futureForecastYear = 3;

        Console.WriteLine(
```

```
$"Forecasted value after {futureForecastYear} years is " +  
FinancialForecasting(pastData[3], futureForecastYear, avgGrowthRate)  
);  
}  
}
```



The screenshot shows a Visual Studio Debug Console window. The title bar reads "Microsoft Visual Studio Debug Console". The output text is as follows:

```
Forecasted value after 3 years is 2200  
C:\Users\HP\source\repos\MyProgram\MyProgram\bin\Debug\net10.0\MyProgram.exe (process 21368) exited with code 0 (0x0).  
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.  
Press any key to close this window . . .
```